OpenShift Ingress

Make your HTTP (or HTTPS) network service available using a protocol-aware configuration mechanism, that understands web concepts like URIs, hostnames, paths, and more. The Ingress concept lets you map traffic to different backends based on rules you define.

Ingress Options	Annotation	DNS	Certificate	Production Example
1) Internal Bank Network - OCP	openshift-core-zone	Automatic (wildcard) 🚀	Automatic Creation and Renewal (Keyfactor) 🚀	myapp.pd-p.**ocp.fcbint.net**
2) Internal Bank Network - Custom	openshift-core-zone	DNS Request	Requesting Internal/External Digital Certificates	myapp**.fcbint.net**
3) External	openshift-secure-zone	DNS Request *and* Akamai WAF Request	Requesting Internal/External Digital Certificates	myapp**.firstcitizens.com**

Option	Details
1) Internal OCP	**Firewall** Submit a Firewall Request for your application if you need access from User Zone. ## DNS The Container Platform team maintains a **wildcard DNS** for each Ingress point. There is no action required for Dev and QA. For UAT and Prod, you can optionally use Floating DNS, which will keep your application DNS the same for both primary and secondary locations, avoiding the need to change URLs. **Certificates** The infrastructure team has deployed Cert Manager for issuance of certificates within our OpenShift clusters. This tool is configured currently to sign certificates using **Keyfactor**. To request certificates, you need only to create an additional manifest within your namespace.

Environment	Application URL	Floating DNS**
Dev	<application>.apps.dv-p.ocp.fcbint.net	N/A
QA	<application>.apps.qa-p.ocp.fcbint.net	N/A
UAT	<application>.apps.ut-p.ocp.fcbint.net <application>.apps.ut-s.ocp.fcbint.net	Submit a DNS Request for a new CNAME record for **<application>.**apps.**ut**.ocp.fcbint.net pointing to the primary Application URL (ut-p). When failing over to the secondary location, another DNS request will be required (ut-s).
Prod	<application>.apps.pd-p.ocp.fcbint.net <application>.apps.pd-s.ocp.fcbint.net	Submit a DNS Request for a new CNAME record for **<application>.**apps.**pd**.ocp.fcbint.net pointing to the primary Application URL (pd-p). When failing over to the secondary location, another DNS request will be required (pd-s).

** *Optiional, but recommended*

```
apiVersion: cert-manager.io/v1
kind: Certificate
metadata:
  annotations:
    metadata.command-issuer.keyfactor.com/Certificate-Support-Group-Contact: MyTeam@firstcitizens.
com
    metadata.command-issuer.keyfactor.com/Environment: Non-Prod-UAT
    metadata.command-issuer.keyfactor.com/Description: "Example Service"
    metadata.command-issuer.keyfactor.com/Organization: FCB
    metadata.command-issuer.keyfactor.com/Application-ID: "12345"
    metadata.command-issuer.keyfactor.com/Platform: OPENSHIFT
  name: example
  namespace: my-example-namespace
spec:
  commonName: example.apps.ut-p.ocp.fcbint.net
  dnsNames:
  - example.apps.ut-p.ocp.fcbint.net
  - example.apps.ut-s.ocp.fcbint.net
  - example.apps.ut.ocp.fcbint.net # Floating DNS
  - example.my-example-namespace.svc.cluster.local # Internal cluster DNS
  duration: 8760h0m0s # One year (default) which is the max allowed
  issuerRef:
    group: command-issuer.keyfactor.com
    kind: ClusterIssuer
    name: keyfactor
  privateKey:
    algorithm: RSA
    size: 4096
    rotationPolicy: Always
  renewBefore: 1080h0m0s # 45 days
  secretName: example-service-tls
  secretTemplate:
    labels:
      app.kubernetes.io/instance: my-example-instance
      app.kubernetes.io/name: my-example-name
      fcbint.net/app-id: "12345"
```

Ingress Example

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  annotations: route.openshift.io/termination: reencrypt ## Recommended (https) (default is edge
/http)
  name: my-ingress
  namespace: my-app-namespace
spec:
  ingressClassName: openshift-core-zone
  rules:
  - host: example.apps.dv-p.ocp.fcbint.net
    http:
      paths:
      - backend:
          service:
            name: service-name
            port:
              name: https
        path: /
        pathType: Prefix
  tls:
  - hosts:
    - example.apps.dv-p.ocp.fcbint.net
    secretName: myapp-tls
```

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  annotations: route.openshift.io/termination: reencrypt ## Recommended (https) (default is edge
/http)
  name: my-ingress
  namespace: my-app-namespace
spec:
  ingressClassName: openshift-core-zone
  rules:
  - host: example.apps.ut.ocp.fcbint.net
    http:
      paths:
      - backend:
          service:
            name: service-name
            port:
              name: https
        path: /
        pathType: Prefix
   - host: example.apps.ut-p.ocp.fcbint.net
    http:
      paths:
      - backend:
          service:
            name: service-name
            port:
              name: https
        path: /
        pathType: Prefix
   - host: example.apps.ut-s.ocp.fcbint.net
    http:
      paths:
      - backend:
          service:
            name: service-name
            port:
              name: https
        path: /
        pathType: Prefix
  tls:
  - hosts:
    - example.apps.ut.ocp.fcbint.net
    - example.apps.ut-p.ocp.fcbint.net
    - example.apps.ut-s.ocp.fcbint.net
    secretName: myapp-tls
```

Restrictions

Our Certificate Signing policies allow the following types of certificates:

1. The `commonName` field must exist, must be less than 64 characters in length, and must contain the appropriate subdomain for the given environment
   a. e.g. `.ut-p.fcbint.net` or `.ut-p.fcbint.net` for UAT primary/secondary clusters
   b. You may also issue `svc.cluster.local` certificates for inner-cluster traffic. This can be used to setup mutual tls authentication for service-to-service communications.
2. The dnsNames field must match a similar subdomain
3. Max Duration must be specified in the format shown above, and cannot exceed 1 year
4. The `privateKey.algorithm` must be RSA
5. The `privateKey.size` must be either 2048 or 4096
6. The issuerRef must be filled out as listed in the example.

Renewal

Cert Manager will automatically renew certificates at their `renewBefore` date. This does away with the need to monitor certificate lifecycle.

If you are only using the certificate within your route objects with a TLS termination policy of `Edge` , this will be 100% non-interrupting for your service.

If, however, you are including this certificate by mounting it as a volume into your pod for mutual tls authentication, passthrough TLS termination, or reencrypt TLS termination, you must ensure the `renewBefore` date is a minimum of 45 days, which aligns with the cluster patching policy that forces all pods to restart (and therefore reload volumes).

Extended Usages

We allow you to select any of the following usages as options for your certificates. Unless any number of usages has been set, cert-manager will set the default requested usages of `digital signature`, `key encipherment`, and `server auth`. Cert-manager will **not** attempt to request a new certificate if the current certificate does not match the current key usage set. To accomplish this, you must delete and recreate the certificate resource.

```
usages:
  - client auth
  - digital signature
  - key agreement
  - key encipherment
  - server auth
```

2) Internal Custom

⚠️

- **Custom DNS** are requested and created in the application namespace and they are **owned, monitored and renewed by the corresponding app team.**
- Infra team **will not keeping track of the custom cert expiry**

Firewall
☑ Submit a Firewall Request for your application if you need access from User Zone.

DNS

☑ Submit an A Record DNS Request for your custom FQDN **<application>.fcbint.net** pointing to the corresponding **Load Balancer IP**:

Environment	Load Balancer	Load Balancer IP
Dev	apps.dv-p.ocp.fcbint.net	10.7.72.209
QA	apps.qa-p.ocp.fcbint.net	10.7.72.212
UAT Primary	apps.ut-p.ocp.fcbint.net	10.7.72.215
UAT Secondary	apps.ut-s.ocp.fcbint.net	10.7.72.218
Prod Primary	apps.pd-p.ocp.fcbint.net	10.0.192.194
Prod Secondary	apps.pd-s.ocp.fcbint.net	10.208.4.34

Certificates

☑ Request a Internal Digital Certificate - initial creation and renewal process

☑ Once a certificate is obtained, this will need to be created as a Secret inside your namespace

☑ **Ingress Example**

Certificate

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: my-ingress
  namespace: my-app-namespace
  labels:
    app.kubernetes.io/component: ingress
    app.kubernetes.io/instance: my-example-instance
    app.kubernetes.io/name: my-example-name
    fcbint.net/app-id: "12345"
spec:
  ingressClassName: openshift-core-zone
  rules:
  - host: example.fcbint.net
    http:
      paths:
      - backend:
          service:
            name: my-example-service
            port:
              name: https
        path: /
        pathType: Prefix
  tls:
  - hosts:
    - example.fcbint.net
    secretName: myapp-tls  # This Secret is also required
```

3) External

Firewall

✅ Submit a Firewall Request for your application URL.

DNS / Akamai

1. Users land on <app>.firstcitizens.com which points to CNAME of <app>.firstcitizens.com.edgekey.net
2. Akamai internally routes the traffic to the Origin for your application <app>-ext.firstcitizens.com, which will point to one of the F5 Public IPs in front of OpenShift

✅ Submit a DNS Request to point the Akamai origin for your external domain (e.g. application.firstcitizens.com) to the corresponding **F5 Public IP.**

- Your Akamai origin will be: **<app>-ext**.firstcitizens.com

Environment	Load Balancer	F5 Public IP
Dev	s-apps.dv-p.ocp.fcbint.net (192.168.62.193)	69.89.135.21
QA	s-apps.qa-p.ocp.fcbint.net (192.168.62.194)	69.89.135.22
UAT Primary	s-apps.ut-p.ocp.fcbint.net (192.168.62.195)	69.89.135.23
UAT Secondary	s-apps.ut-s.ocp.fcbint.net (192.168.62.196)	69.89.135.24
Prod Primary	s-apps.pd-p.ocp.fcbint.net (172.20.27.178)	69.89.129.25
Prod Secondary	s-apps.pd-s.ocp.fcbint.net (172.27.87.13)	69.89.158.23

✅ Submit an Akamai CDN/Kona (WAF) policy configuration Request to have your domain protected by the Akamai web application firewall.

- Include your <application>.firstcitizens.com and the origin: <application>-ext.firstcitizens.com

✅ Submit a DNS request for your <app>firstcitizens.com to point to the Akamai CNAME

- <app>.firstcitizens.com <app>.firstcitizens.com.edgekey.net

Certificates

- ✅ Request a External Digital Certificate - initial creation and renewal process
- ✅ Once a certificate is obtained, this will need to be created as a Secret inside your namespace

✅ **Ingress Example**

Certificate

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: my-ingress
  namespace: my-app-namespace
  labels:
    app.kubernetes.io/component: ingress
    app.kubernetes.io/instance: my-example-instance
    app.kubernetes.io/name: my-example-name
    fcbint.net/app-id: "12345"
spec:
  ingressClassName: openshift-secure-zone
  rules:
  - host: example.firstcitizens.com
    http:
      paths:
      - backend:
          service:
            name: my-example-service
            port:
              name: https
        path: /
        pathType: Prefix
  tls:
  - hosts:
    - example.firstcitizens.com
    secretName: myapp-tls  # This Secret is also required
```

Monitoring

TODO: Dynatrace dashboard for certificates per namespace

Additionally, you can always run `oc get certificate` or `oc describe certificate` from the CLI.

For more details see: OpenShift Cert Manager

OpenShift HAProxy Ingress Annotations

Summary

The following describes HAProxy Ingress annotations supported by OpenShift.

Annotation	Description
`haproxy.router.openshift.io/balance`	Sets the load-balancing algorithm. Available options are `random`, `source`, `roundrobin`, and `leastconn`. The default value is `source` for TLS passthrough routes. For all other routes, the default is `random`.
`haproxy.router.openshift.io/pod-concurrent-connections`	Sets the maximum number of connections that are allowed to a backing pod from a router. Note: If there are multiple pods, each can have this many connections. If you have multiple routers, there is no coordination among them, each may connect this many times. If not set, or set to 0, there is no limit.
`haproxy.router.openshift.io/rate-limit-connections`	Setting `'true'` or `'TRUE'` enables rate limiting functionality which is implemented through stick-tables on the specific backend per route. Note: Using this annotation provides basic protection against denial-of-service attacks.
`haproxy.router.openshift.io/rate-limit-connections.rate-http`	Limits the rate at which a client with the same source IP address can make HTTP requests. It accepts a numeric value. Note: Using this annotation provides basic protection against denial-of-service attacks.

`haproxy.router.openshift.io/timeout`	Sets a server-side timeout for the route. (TimeUnits) Supported time units are microseconds (us), milliseconds (ms), seconds (s), minutes (m), hours (h), or days (d). Example value: "60s"
`haproxy.router.openshift.io/ip_whitelist`	Sets an allowlist for the route. The allowlist is a space-separated list of IP addresses and CIDR ranges for the approved source addresses. Requests from IP addresses that are not in the allowlist are dropped. The maximum number of IP addresses and CIDR ranges directly visible in the `haproxy.config` file is 61. [1]
`haproxy.router.openshift.io/rewrite-target`	Sets the rewrite path of the request on the backend.
`haproxy.router.openshift.io/set-forwarded-headers`	Sets the policy for handling the `Forwarded` and `X-Forwarded-For` HTTP headers per route. The values are: `append`: appends the header, preserving any existing header. This is the default value. `replace`: sets the header, removing any existing header. `never`: never sets the header, but preserves any existing header. `if-none`: sets the header if it is not already set.